

Topological spaces

Definition: Let X be a nonempty set. A class $\mathcal{T}$ of subsets of X is called a topology on X if $\mathcal{T}$ satisfies the following 3 axioms:

$$(T1) \quad \emptyset, X \in \mathcal{T}$$

$$(T2) \quad A_1, A_2 \in \mathcal{T} \Rightarrow A_1 \cap A_2 \in \mathcal{T} \quad ..$$

$$(T3) \quad A_i \in \mathcal{T} \text{ for } i \in I \Rightarrow \bigcup_{i \in I} A_i \in \mathcal{T}.$$

The members of $\mathcal{T}$ are called $\mathcal{T}$ -open sets or simply open sets & the ordered pair $(X, \mathcal{T})$ is called a topological space.

Definition: Let $(X, \mathcal{T})$ be a topological space. A subset K of X is called $\mathcal{T}$ -closed or simply closed if its complement $K^I = K^c = X \setminus K$ of K in X is a $\mathcal{T}$ -open set. In

other words K is τ -closed $\Leftrightarrow$
 $X \setminus K \in \tau$.

Theorem: Let (X, τ) be a topological space and τ_K be the class of τ -closed subsets of X . Then τ_K possesses the following properties:

$$(C1) \quad \emptyset, X \in \tau_K$$

$$(C2) \quad K_1, K_2 \in \tau_K \Rightarrow K_1 \cup K_2 \in \tau_K$$

$$(C3) \quad K_i \in \tau_K \text{ for } i \in I \Rightarrow \bigcap_{i \in I} K_i \in \tau_K.$$

Conversely if a nonempty class τ_K of subsets of X satisfies (C1), (C2) & (C3) then

$$\tau = \{A \mid A \subseteq X, X \setminus A \in \tau_K\}$$

is a topology on X & the

members of τ_K are τ -closed subsets of X .

Proof:

$$(C1) \quad \emptyset' = X \in \tau \text{ by (T1)} \Rightarrow \emptyset \in \tau_K.$$

$$X' = \emptyset \in \tau \text{ by (T1)} \Rightarrow X \in \tau_K.$$

(C2) Let $K_1, K_2 \in \mathcal{T}_K$

$$(K_1 \cup K_2)' = \underbrace{(K_1') \cap (K_2')}_{\in \mathcal{T} \text{ by (T2)}} \stackrel{\in \mathcal{T}}{\text{by de Morgan's Law}}$$

$$\Rightarrow K_1 \cup K_2 \in \mathcal{T}_K$$

(C3) Let $K_i \in \mathcal{T}_K$ for $i \in I$.

$$\left(\bigcap_{i \in I} K_i \right)' = \underbrace{\bigcup_{i \in I} (K_i')}_{\in \mathcal{T} \text{ by (T3)}} \stackrel{\in \mathcal{T}}{\text{by de Morgan's Law.}}$$

$$\Rightarrow \bigcap_{i \in I} K_i \in \mathcal{T}_K$$

Let $\mathcal{T}_K$ be a nonempty class of subsets of X satisfying (C1), (C2) & (C3).

We shall show that

$$\mathcal{T} = \{ A \mid A \subseteq X, A' \in \mathcal{T}_K \}$$

is a topology on X .

$$(T1) \quad \emptyset' = X \in \mathcal{T}_K \quad \text{by (C1)} \Rightarrow \emptyset \in \mathcal{T} \\ X' = \emptyset \in \mathcal{T}_K \quad \text{by (C1)} \Rightarrow X \in \mathcal{T}.$$

(T2) Let $A_1, A_2 \in \mathcal{T}$. Then

$$A_1', A_2' \in \mathcal{T}_K. \quad \text{By (C2)} \quad A_1' \cup A_2' \in \mathcal{T}_K.$$

But

$$A_1' \cup A_2' = (A_1 \cap A_2)' \quad \text{by de Morgan's Law.}$$

$$\text{Thus} \quad A_1 \cap A_2 \in \mathcal{T}.$$

(T3) Let $A_i \in \mathcal{T}$ for $i \in I$.

Then $A_i' \in \mathcal{T}_K$ for $i \in I$. By (C3)

$$\bigcap_{i \in I} A_i' \in \mathcal{T}_K. \quad \text{But} \quad \bigcap_{i \in I} A_i' = \left(\bigcup_{i \in I} A_i \right)'$$

by de Morgan's Law. So

$$\left(\bigcup_{i \in I} A_i \right)' \in \mathcal{T}_K. \Rightarrow \bigcup_{i \in I} A_i \in \mathcal{T}.$$

$\therefore (X, \tau)$ is a topological space.

Let $K \subseteq X$. Now

K is τ -closed $\iff K' \in \tau$

$$\iff (K')' = K \in \tau_K$$

Examples:

1. Let $X \neq \emptyset$ and $\mathcal{D}$ be the class of all subsets of X . But we know that the class consisting of all subsets of X is the power set $\mathcal{P}(X)$ of X . Here we denote $\mathcal{P}(X)$ by $\mathcal{D}$. In this case $\mathcal{D}$ is a topology on X . This topology is called the discrete topology on X . $(X, \mathcal{D})$ is the discrete topological space or simply the discrete space.

In this topology any subset of X is $\mathcal{D}$ -open. Consider the class

$\mathcal{D}_K$ of $\mathcal{D}$ -closed subsets of X .

We know that $\mathcal{D}_K$ is obtained from $\mathcal{D}$ by finding the complement of each set in $\mathcal{D}$. Thus $\mathcal{D}_K = \mathcal{D}$, i.e. any subset of X is both $\mathcal{D}$ -open & $\mathcal{D}$ -closed in discrete topological space.

2. Let $X \neq \emptyset$. Assume that $\tau = \{\emptyset, X\}$. Then (X, τ) is a topological space. It is called the indiscrete topological space or the trivial space. (τ is called the indiscrete topology or the trivial topology on X). In this space $\emptyset$ & X are the only subsets of X which are both open & closed.

3. Let $X = \{a, b, c, d, e\}$ & $\tau = \{\emptyset, X, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}\}$.

(i) Show that (X, τ) is a topological space. (Exercise)

(ii) Find all τ -closed subsets of X .

(iii) If any find all nonempty, proper subsets of X which are both open & closed.

(iv) Is there any subset of X which

is neither open nor closed?

Solution:

(ii) Recall that $A \subseteq X$ is closed
or equivalently $A \in \mathcal{C}_K$ iff $X \setminus A \in \mathcal{C}$.
 $\mathcal{C}_K = \{X, \emptyset, \{b, c, d, e\}, \{a, b, e\}, \{b, e\}, \{a\}\}$

(iii) $\{a\}, \{b, c, d, e\}$ are the only
subsets of X which are both open &
closed.

(iv) $\{b, d\} \subseteq X$ but
 $\{b, d\} \notin \mathcal{C} \cup \mathcal{C}_K$. Thus $\{b, d\}$ is
neither open nor closed.

4. Let $X \neq \emptyset$ and
 $\mathcal{C} = \{\emptyset\} \cup \{A \mid A \subseteq X, A^c \text{ is finite}\}$.
Then $\mathcal{C}$ is a topology on X . (It is
called the cofinite topology or the
finite complement topology on X .)
Show it.

Solution:

(T1) By definition of $\mathcal{T}$, $\emptyset \in \mathcal{T}$.

Since $X \setminus X = \emptyset$ & $\emptyset$ is finite
 $X \in \mathcal{T}$.

(T2) Let $A_1, A_2 \in \mathcal{T}$.

- $A_1 = \emptyset$ or $A_2 = \emptyset$. In this case $A_1 \cap A_2 = \emptyset \in \mathcal{T}$.

- $A_1 \neq \emptyset$ & $A_2 \neq \emptyset$. Then A_1', A_2' are finite set and so $(A_1 \cap A_2)' = A_1' \cup A_2'$ is finite.

Thus $A_1 \cap A_2 \in \mathcal{T}$.

(T3) Let $A_i \in \mathcal{T}$ for each $i \in I$.

- If $A_i = \emptyset$ for each $i \in I$ then $\bigcup_{i \in I} A_i = \emptyset \in \mathcal{T}$.

- Assume that there exists an $i_0 \in I$ s.t. $A_{i_0} \neq \emptyset$.